

Lab 4 - Vibe Coding Conversation

Arjun Srikanth | PES1UG24AM920 | Section H

Assigned game: SETAPESU26/08_helicopter | AI assistant: Codex

7 October 2026

Editorial note: Wording has been formally edited and related assistant updates consolidated from the recorded Lab 4 discussion.

1. Assignment request and implementation

Student

Please complete Lab 4 using the supplied handout and maintain a state file so that the project's progress and next steps remain available.

AI assistant

I identified SETAPESU26/08_helicopter as the assigned starter for PES1UG24AM920. The required work consists of a movement and boundary fix, obstacle collision with game over, distance scoring, and a visible one-hit shield. I preserved the unchanged starter and recorded the before video prior to implementation. Each required task was committed separately.

AI assistant

The movement fix caps vertical speed, responds promptly when the direction changes, and contains the complete player rectangle within both screen boundaries. Wall contact produces game over, while passage through a clear gap remains safe. Distance increases during play, remains visible after game over, and resets on restart.

AI assistant

The initial shield implementation tracked continued contact with an obstacle after protection was consumed. Fourteen behavioral tests passed at that stage. Subsequent gameplay feedback identified a weakness in that contact model, which was corrected and then simplified as described below. These initial results are historical, not a description of the final shield implementation.

2. Conversation publication preference

AI assistant

The starter also requires a conversation-page link. Should the submission use a Lab 4-only transcript on GitHub or a public share link to the entire conversation?

Student

Use a Lab 4-only transcript on GitHub.

3. Review of the video deliverables

Student

Please confirm that all instructions and deliverables have been completed. I have not seen the video deliverables, so please verify their availability.

AI assistant

The videos had been recorded and uploaded, but they had not been displayed in the earlier handoff. I reopened both recordings, checked playback and duration, verified that the local files matched the GitHub copies, and displayed the before and after videos for review.

4. Gameplay feedback and collision correction

Student

The after version still appears to have issues: the player moves downward out of view and passes through walls. Please review both the video and the working game as often as necessary to ensure that the result looks correct and functions properly.

AI assistant

I found that the initial shield model allowed continued overlap with a visibly intact wall after protection had been consumed. This made the collision behavior appear incorrect. The first correction cleared the struck wall immediately and kept later solid-wall contact lethal. I also moved the HUD below the playfield so that it would not obscure the player near the bottom boundary.

AI assistant

The behavioral suite was expanded to 18 tests, including exact wall contact, safe gap positions, long-held movement keys, shield consumption, score reset, and restart. Revised native recordings were inspected for the boundary, shield-hit, and unprotected-collision transitions. The independent simulations reported at this stage concerned that intermediate implementation; they are not represented as new tests of the later simplified version.

5. Simplification and before/after comparison

Student

The implementation appears unnecessarily complex for this simple game, and the work is taking longer than expected. Please return to the assigned deliverables, fix the actual issues with a proportionate solution, and allow me to play both the original and completed versions.

AI assistant

I removed the additional per-wall state and hit timer. In the final implementation, an absorbed hit consumes the shield and removes the hit obstacle. The next unprotected obstacle collision ends the game. This removed 33 runtime lines from the preceding revision and restored the obstacle class to the starter implementation.

AI assistant

All 18 behavioral tests passed for the simplified game. A matching ten-second after video demonstrated bottom containment, prompt upward reversal, safe gap passage, one absorbed shield hit, a later unprotected crash, and restart. Both native versions were launched for manual comparison. Up/Down controls movement; in the completed version, Space activates the shield and R restarts after game over.

6. Confirmation of the assigned game and shield

Student

Please confirm that this is the game assigned to me. I cannot see a helicopter image and would like to understand the required final deliverables. Explain the shield, whether it should remain, and update the submission files accordingly.

AI assistant

The official Section H assignment sheet maps PES1UG24AM920 to game 8, SETAPESU26/08_helicopter. The original renderer draws the player as a dark rounded rectangle; no helicopter sprite is supplied. Retaining that appearance follows the starter. The shield is explicitly required by Task 4: visible activation, one absorbed obstacle collision, and immediate loss of protection after use. Removing the hit obstacle is an implementation choice, not an additional instructor requirement.

AI assistant

The required package contains updated code, genuine ten-second before and after videos, and a conversation document under Lab-4. The starter additionally requests a conversation-page link. The four tasks have separate commits. No pull request is made to SETAPESU26, and the accepted Labs 1-3 remain unchanged.

7. Final verification

Student

Please check once more whether the videos and deliverables meet the requirements.

AI assistant

Both MP4s decoded without errors and measured exactly 10.000 seconds at 30 frames per second, with 300 frames each. The game passed all 18 behavioral tests. The published source, videos, PDF, and README links matched the final local package.

Submission references

Personal repository

https://github.com/arjunsrikanthh/PES1UG24AM920_se_lab/tree/main/Lab-4

Assigned starter and task specification

https://github.com/SETAPESU26/08_helicopter

Conversation page

https://github.com/arjunsrikanthh/PES1UG24AM920_se_lab/blob/main/Lab-4/PES1UG24AM920.pdf